Analysis Setup

March 27, 2026

1 Spectral Class:

The near sample shows an extremely dense concentration of M dwarfs and the far sample shows this region much less populated.
Conversely, the far sample shows a slightly larger proportion of luminous stars (K, G, F, A types) relative to the near sample. These are intrinsically bright enough to be seen at large distances, and the far sample volume is large enough to contain many of them.

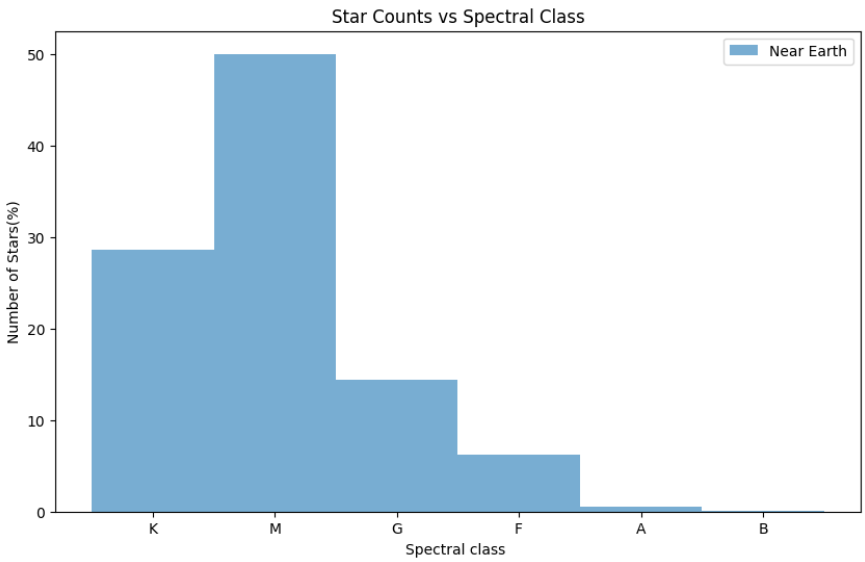

Figure 1: Spectral class for near 50k.

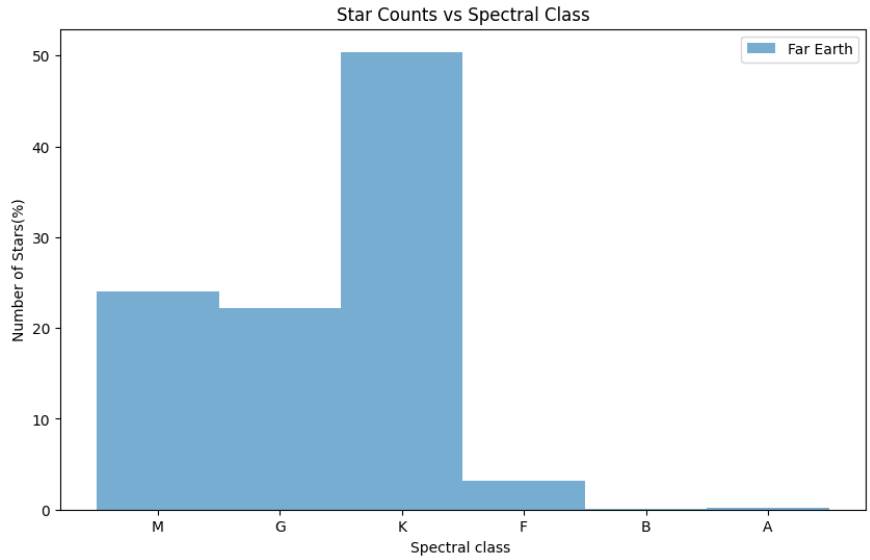

Figure 2: Spectral Class for far 50k.

The shift from M-dominated to K/G-dominated between the near and far samples is entirely a survey selection effect, not a physical difference in the stellar populations along these lines of sight. The true stellar IMF predicts that M dwarfs should be the most numerous stars everywhere in the Galaxy by a large margin — typically comprising 70–80 percent of all stars by number. The far sample's reduced M dwarf fraction reflects simply that most M dwarfs are too faint to detect at large distances.

2 Luminosity

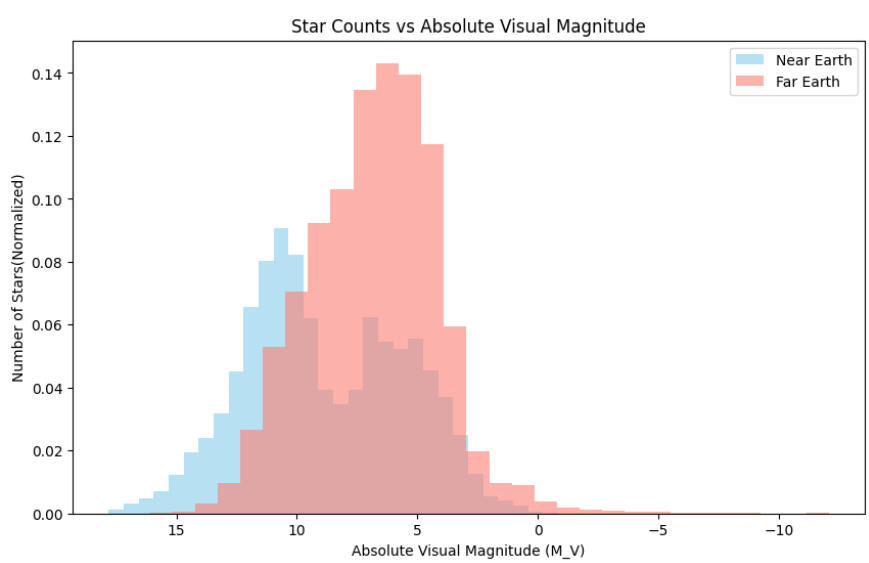

Figure 3: Number of Stars vs. Absolute visual magnitude.

In the near sample, the count rises steeply toward faint magnitudes, peaking in the M dwarf regime ($M_V \sim 11$–13).

This peak directly traces the IMF (a fundamental astrophysical probability distribution function that describes the relative number of stars of different masses produced in a single star-formation event): low-mass stars are the most numerous type of star in the Galaxy.

The far sample's luminosity function rises toward intermediate magnitudes ($M_V \sim 5$–8, corresponding to K and G dwarfs) but drops off sharply at $M_V > 10$. This cutoff is the Gaia magnitude limit in action. Stars fainter than $M_V \sim 10$ at distances of several hundred parsecs have apparent magnitudes below Gaia's photometric completeness threshold and simply do not appear in the catalogue.

3 Effective Temperature:

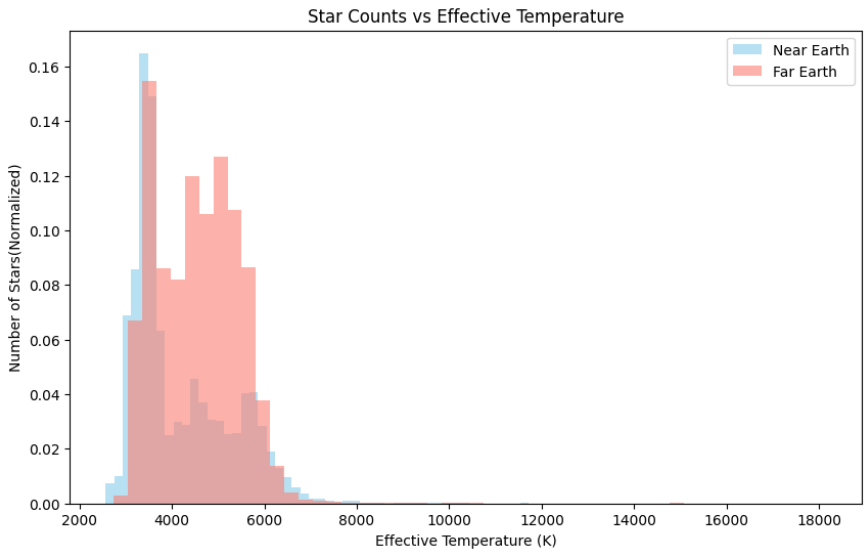

Figure 4: Number of Stars vs. Effective Temperature.

The effective temperature distribution directly translates the spectral class distribution into temperature space. The near sample shows a cool-peaked distribution with the maximum around 3400–3800 K (mid-M dwarfs). The far sample peaks around 4500–5000 K (late K dwarfs).

The mean effective temperature of the near sample is approximately 4204.77 K while the far sample mean is approximately 4553.92 K.

4 Main Sequence:

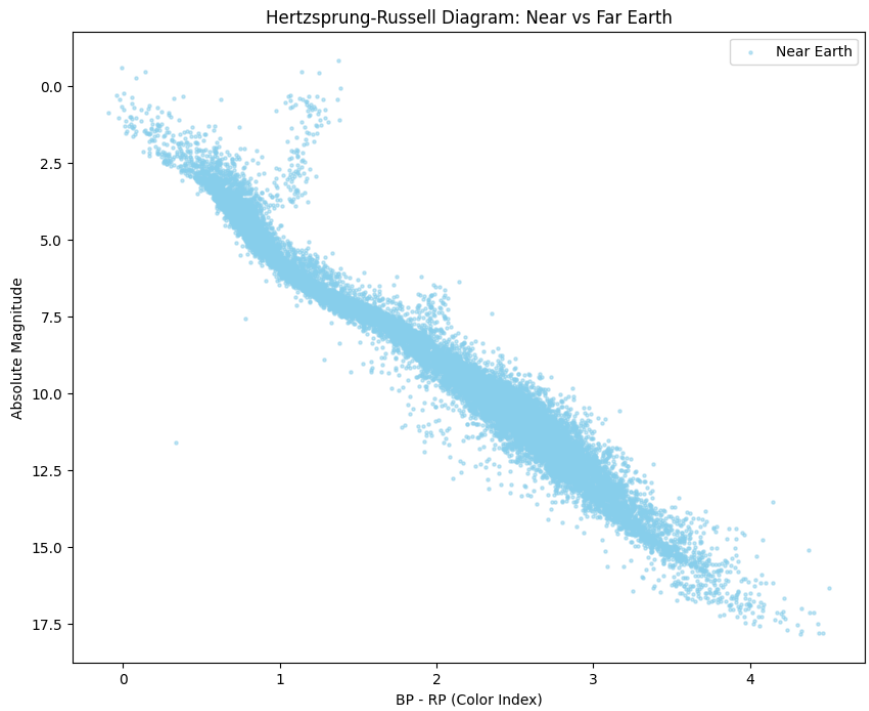

Figure 5: $M_V vs. Color Index$.

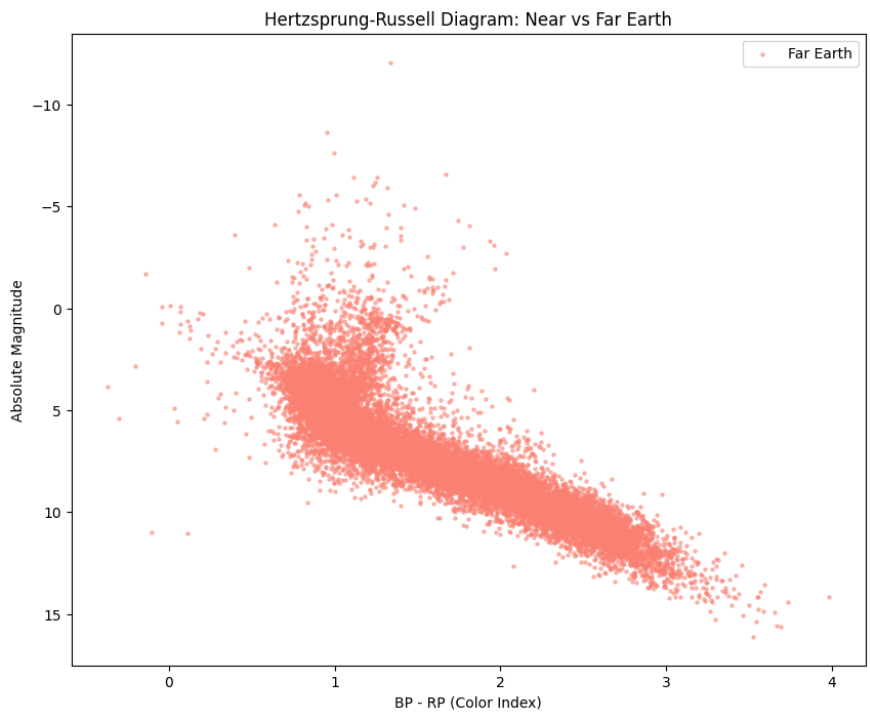

Figure 6: $M_V vs. Color Index$,

The Main sequence of both samples is broad due the sample being a field star sample.

Both HR diagrams show the characteristic morphology of a mixed field star population: a broad, scattered main sequence dominated by low-mass stars, with no clean evolutionary turn-off point and no coherent giant branch. The main sequence dominance in both samples reflects stellar lifetimes, not stellar age — low-mass stars spend billions of years on the main sequence, making them the statistically most likely type of star to be observed at any random moment.

5 Conclusion

The HR diagram analysis of the Gaia DR3 near and far field star samples yields the following principal results:

1. Both samples are dominated by main sequence stars, consistent with the expectation that stars spend the majority of their lives in core hydrogen burning. This dominance reflects stellar lifetimes, not a common formation event.

2. The near sample (< 100 pc) is heavily dominated by M dwarfs , which are detectable at these distances. The far sample (shows a reduced M dwarf fraction due to Gaia's magnitude limit excluding faint stars at large distances — a survey selection effect, not a physical difference.

3. The mean effective temperature is higher in the far sample (4,553.92 K) than the near sample (4,204.77K) for the same reason: the magnitude limit biases the far sample toward intrinsically hotter, more luminous stars.

4. Both main sequences are broad and scattered, arising from the superposition of stars with different distances, reddening values, metallicities, and binary fractions.

6 Note:

This is a very Rough outline of the analysis and inferences to be done for the project. This example analysis used the near earth clean calc.csv and the far earth 50k clean calc.csv. Since this data is a field star data, isochrone fitting cannot be used as the stars observed are not from the same formation time (did not form at the same time).
This is just an example for what the analysis part of the report will cover.
Need to add references, choose better range-colorscheme for the plots, and plot for the other datasets.
There are some more methods that I'm looking into but this is mainly what the analysis will cover.